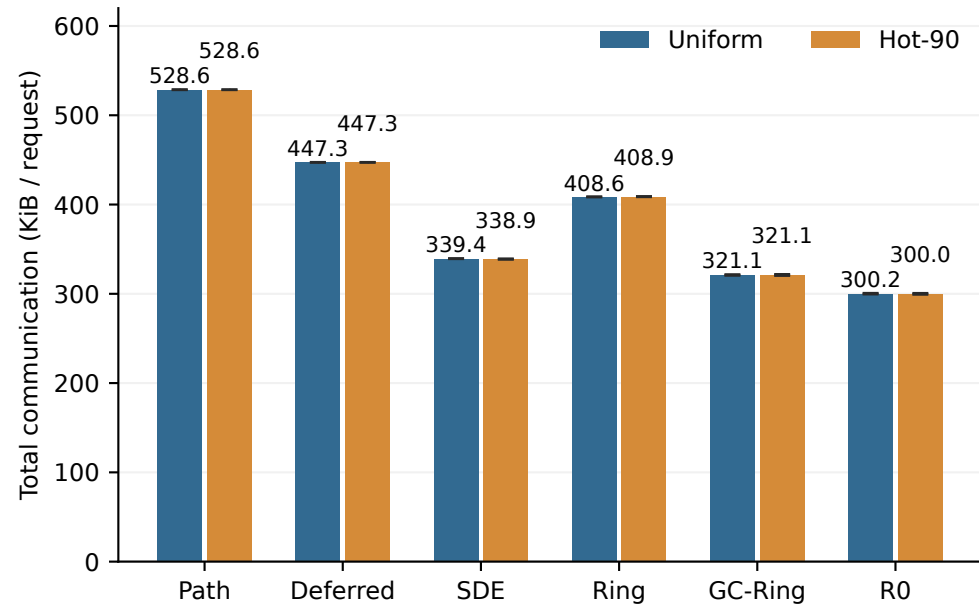
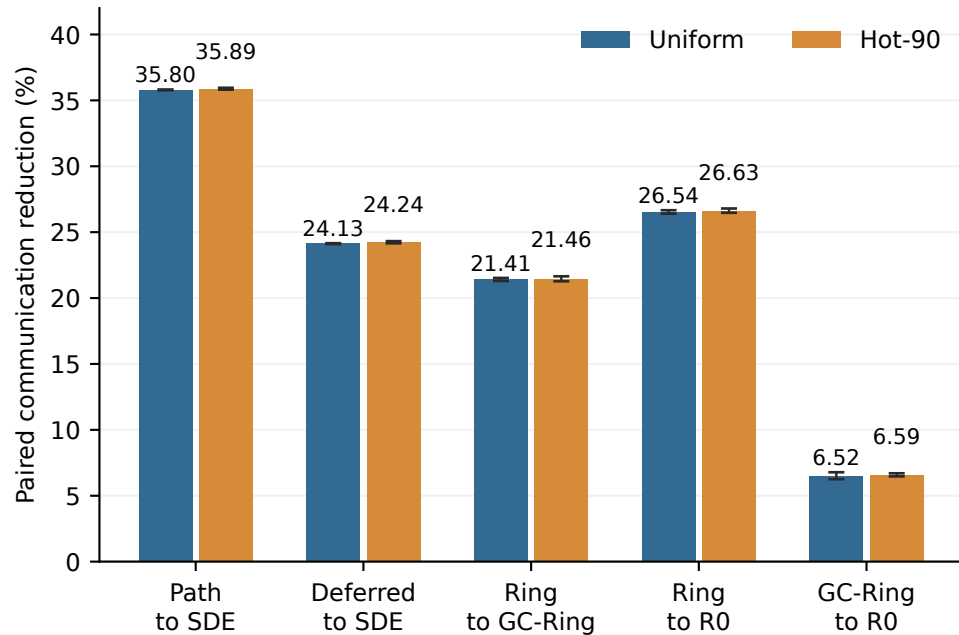


(a) Absolute bandwidth cost



(b) Explicit baseline comparisons



N = 16,384; B = 4,096 bytes; fixed Z4/A3/S3; five paired runs, mean and 95% t intervals.
 2,048 warmup + 4,096 measured requests. Includes headers, nonces, authentication and recursive maps.
 Finite measurement window; same-profile comparison, not the separately tuned result. No latency claim.